EcoQuest — Quick Revision Sheet (Viva Cheat Sheet) ⚡

B.Sc. Computer Science Community Engagement Project (CEP)
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Academic Year: 2026–2027 | **Guide:** `[Guide Name]` | **Roll No:** `[Roll Number]`
Keep this sheet for 5-minute rapid review immediately before entering the viva room.

1. Project Introduction (30-Second Elevator Pitch)

- **Title:** EcoQuest — Campus Restoration Game (Theme: Gamified App for Sustainability).
- **Core Problem:** Campus sustainability guidelines (saving energy, segregating waste, saving water) suffer from passive awareness and low engagement because individual effort feels invisible ("drop in the ocean").
- **Solution:** A mobile app that gamifies eco-habits. Students complete daily micro-quests to earn **Lifetime Points** (for levels and rank) and build **Spendable Funds** to restore a virtual campus quad through four sequential physical upgrades.
- **Key Innovation:** Two-tier economy where spending points to upgrade the campus never lowers player level or leaderboard standing.

2. Technology Stack At a Glance

- **Framework:** Expo SDK `~57.0.30` (Managed workflow).
- **Mobile Runtime:** React Native `0.86.3` (targeting Android phones via Expo Go).
- **Core Library:** React `19.2.3` (Functional components & hooks: `useState`, `useEffect`, `useRef`, `useContext`).
- **State Management:** React Context (`GameContext.js`) with custom `useGame()` hook.
- **Local Persistence:** `@react-native-async-storage/async-storage` (`2.2.0`) under key `'ecoquest.gameState.v1'`.
- **Safe Area Handling:** `react-native-safe-area-context` (`~5.7.0`) for Android 3-button and gesture navigation clearance.
- **Testing:** Custom automated integration suite (`tests/test_integration.mjs`) with **83/83 tests passing (100%)**.

3. The Five UI Navigation Tabs

1. 🏠 **Home:** Features illustrated `CampusScene` (Stages 0–4), level progress card, points stats, and sequential upgrade action card.
2. 🎯 **Quests:** 8 sustainability challenges across 5 categories with category filters (All, Energy, Waste, Water, Transport, Food) and instant completion buttons.
3. 🏆 **Rank:** Dynamic 8-participant campus leaderboard (7 fictional peers + user) with live rank calculation and Top-3 podium (🥇 Maya, 🥈 Aarav, 🥉 Priya).
4. 🎖️ **Badges:** 8 achievement trophies with live unlock counters (4 milestone badges + 4 category-specific badges).
5. 👤 **Profile:** Demonstrator identity card, cumulative performance summary, and a two-step confirmation **Reset Demo Data** button.

4. Eight Quests & Four Campus Upgrades

8 Quests Catalog (Total Available = 210 Lifetime Points)

ID	Quest Title	Category	Pts	Key Description
`ch-1`	Reusable Water Bottle	Waste ♻️	**30**	Carry a refillable bottle to class.
`ch-2`	Lights & Fans Off	Energy ⚡	**20**	Switch off lights/fans in empty rooms.
`ch-3`	Green Commute	Transport 🚲	**40**	Walk, cycle, or take campus shuttle.
`ch-4`	Plant-Based Meal	Food 🥗	**25**	Eat vegetarian lunch at cafeteria.
`ch-5`	Segregate Campus Waste	Waste ♻️	**35**	Separate wet and dry waste into bins.
`ch-6`	5-Minute Shower	Water 💧	**25**	Limit dorm shower time to < 5 min.
`ch-7`	Unplug Idle Chargers	Energy ⚡	**15**	Eliminate phantom power drain.
`ch-8`	Bring a Tote Bag	Waste ♻️	**20**	Use cloth bag for campus bookstore.

4 Sequential Campus Upgrades (Total Cost = 180 Points)

- **Stage 1 — Native Shade Trees (30 pts):** Restores lawn grass and biodiversity in central quad.
- **Stage 2 — Rooftop Solar Array (40 pts):** Photovoltaic solar panels on Academic Hall roof.
- **Stage 3 — Smart Compost Hubs (50 pts):** Color-coded compost pods along campus walkways.
- **Stage 4 — Pure Water Cistern (60 pts):** Chilled water refill fountain in campus plaza.

5. Key Point Totals & Mathematical Invariants

```
Total Available Quest Points = 30 + 20 + 40 + 25 + 35 + 25 + 15 + 20 = 210 pts
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```
Total Upgrade Costs = 30 + 40 + 50 + 60 = 180 pts
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```
Final Spendable Surplus = 210 - 180 = 30 pts remaining
```

- **Invariant 1:** $\text{Lifetime Points} \ge \text{Spendable Points}$ at all times.
- **Invariant 2:** Lifetime Points **never decreases**; it strictly sums completed quest rewards.
- **Invariant 3:** Spendable Points = $\max(0,\ \text{Lifetime} - \text{Total Spent})$. It never falls below 0.
- **Invariant 4:** Upgrades strictly require sequential purchase (Stage 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4).

6. Level & Badge Thresholds

Player Levels (Derived strictly from Lifetime Points)

- **Level 1: Eco Seedling** (0–49 pts)
- **Level 2: Green Scout** (50–99 pts) — *Promoted at 50 pts*
- **Level 3: Earth Guardian** (100–169 pts) — *Promoted at 100 pts*
- **Level 4: Planet Champion** (170–210 pts) — *Promoted at 170 pts; Max completion at 210 pts*

Achievement Badges (8 Trophies)

- `bdg-first` (**Eco Starter** 🌱): Complete 1 quest.
- `bdg-100` (**Green Sprout** 🌿): Reach 100 Lifetime Points.
- `bdg-160` (**Earth Guardian** 🛡️): Reach 160 Lifetime Points *(distinct from Level 4 at 170 pts)*.
- `bdg-200` (**Planet Hero** 🌎): Reach 200 Lifetime Points.
- `bdg-waste` (**Zero-Waste Pioneer** ♻️): Complete any 2 Waste quests (`ch-1` , `ch-5` , `ch-8`).
- `bdg-energy` (**Watts Saver** ⚡): Complete both Energy quests (`ch-2` AND `ch-7`).
- `bdg-commute` (**Clean Cruiser** 🚲): Complete Transit quest (`ch-3`).
- `bdg-water` (**Hydro Guardian** 💧): Complete Water quest (`ch-6`).

7. State Management & Storage Architecture

- **Single Source of Truth:** We persist **only primary ground-truth records**: `completedQuestIds` and `purchasedUpgradeIds` .
- **Pure Selectors:** Lifetime points, spendable balance, stage, level info, badges, and leaderboard are computed dynamically on the fly via pure selector functions.
- **AsyncStorage Hydration:** On startup, `hydrate()` parses and validates JSON under `'ecoquest.gameState.v1'` . A loading card with `ActivityIndicator` displays until hydration finishes.
- **Race-Condition Protection:** Writes are queued sequentially through `saveChainRef` (Promise chain write mutex) alongside synchronous React `useRef` trackers.
- **Two-Step Demo Reset:** `resetDemo()` clears AsyncStorage key, empties arrays, and instantly resets app to Stage 0, Level 1, 0 points, and Rank #8.

8. Five Explicit Project Limitations (Memorize These!)

1. Self-Reported Actions: Quests are completed via simulated button clicks; no IoT sensors, cameras, or GPS geofencing verify real-world behavior.
2. Simulated Leaderboard: Competitor peers (Maya, Aarav, Priya, etc.) are fixed benchmark profiles to test dynamic sorting offline without needing cloud servers.
3. Local-Only Storage: Progress is stored on the physical device via AsyncStorage; no multi-device account sync or cloud database.
4. Pedagogical Impact Figures: Carbon, water, and waste reduction figures are educational demonstration benchmarks, not certified lab measurements.
5. Demonstration Scope: Developed as an academic prototype for college CEP evaluation rather than a commercial app store release.

9. Top 10 Questions Most Likely to be Asked in Viva

1. What is the difference between Lifetime Points and Spendable Points?

Answer: Lifetime Points is the cumulative total of all points earned; it never decreases and determines level, badges, and rank. Spendable Points is available currency deducted when funding campus upgrades.

2. If a student spends points on a campus upgrade, do they lose their level or drop on the leaderboard?

Answer: No. Player level and leaderboard rank are derived strictly from Lifetime Points, completely isolated from Spendable Points.

3. How does the dynamic leaderboard handle equal scores (tie-breaking)?

Answer: Deterministic tie-breaking uses original roster seniority (`initialIndex`). At 210 points, the user ties with Priya Nair (210 pts). Because Priya was registered earlier at index 2 vs. the user's index 7, Priya keeps Rank #3 (Bronze podium) and the user takes Rank #4.

4. What happens if the phone loses internet connection during the demo?

Answer: EcoQuest runs 100% offline. It has zero cloud dependencies and saves everything locally via AsyncStorage, so it works perfectly without internet.

5. Why did you not use a backend database like Firebase or MySQL?

Answer: For an offline college demonstration, a backend introduces network failure points, hosting costs, and latency. AsyncStorage provides instant, robust client-side persistence without external failure risks.

6. What prevents someone from clicking a quest multiple times?

Answer: Three layers of protection: UI buttons disable with a checkmark, `processQuestCompletion()` checks `completedQuestIds.includes(questId)` and rejects duplicates, and synchronous `useRef` checks block rapid double-tapping.

7. What happens if corrupt data is found in AsyncStorage?

Answer: `validateSavedGameState()` wraps parsing in a `try...catch` block. If corrupt, it discards the bad data and safely initializes a fresh, clean demo session without crashing.

8. Why is the Earth Guardian badge at 160 points while Level 4 is at 170 points?

Answer: Level 3 spans 100 to 169 points (titled "Earth Guardian"). The Earth Guardian badge is an advanced milestone trophy awarded at 160 points to reward deep dedication right before breaking into Level 4 (Planet Champion at 170 points).

9. What would be your first step to turn this into a live college-wide system?

Answer: Introduce college email authentication (SSO) and connect `GameContext.js` to a secure backend REST API with PostgreSQL to synchronize student progress to the cloud.

10. How did you verify that all these calculations and game rules work correctly?

Answer: We wrote an automated integration test suite in `tests/test_integration.mjs` containing 83 assertions covering economy balance, levels, badges, leaderboard, persistence, and reset. All 83 tests pass with 100% success rate.
